Global Weather Forecasting Assessment

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1 Introduction

Weather forecasting plays a critical role in agriculture, transportation, disaster preparedness, aviation, environmental planning and day-to-day decision-making. Accurate forecasting models help governments, businesses and individuals respond effectively to changing atmospheric conditions while reducing uncertainty and operational risk.

The objective of this project is to analyze a large-scale global weather dataset and develop machine learning models capable of forecasting future temperature trends. In addition to predictive modeling, the project explores climate patterns, air quality relationships, geographical weather variations and anomaly detection to generate meaningful insights from real-world weather data.

This assessment was completed as part of the PM Accelerator internship evaluation process. PM Accelerator's mission is to empower future product, AI and technology leaders through hands-on learning experiences, real-world projects and career acceleration opportunities that help candidates build impactful solutions and grow professionally.

The project follows a complete data science workflow consisting of data cleaning, exploratory data analysis, feature engineering, model benchmarking, ensemble forecasting and advanced analytical interpretation. The final outcome demonstrates both technical forecasting capability and business-oriented analytical thinking.

2 Dataset Description

The dataset used in this project is the **Global Weather Repository** obtained from Kaggle. It contains daily weather observations collected from cities around the world, providing a large and diverse dataset for climate analysis and forecasting tasks. The dataset includes multiple weather-related variables measured across different countries and geographic locations.

Key variables available in the dataset include temperature, humidity, wind speed, atmospheric pressure, precipitation, air quality indicators, latitude, longitude and country-level location information. In addition, the `last_updated` timestamp column enables time-based trend analysis and forecasting.

This dataset is well-suited for machine learning because it combines temporal, environmental and geographical information in a single source. The variety of features allows both predictive modeling and broader analytical tasks such as anomaly detection, climate comparison and regional weather pattern analysis.

3 Basic Assessment

3.1 Data Cleaning and Preprocessing

Real-world weather datasets often contain inconsistencies such as missing values, duplicate records, outliers and mixed data formats. Therefore, an extensive preprocessing pipeline was applied to improve data quality before performing analysis and model development.

Missing numerical values were handled using median imputation, while missing categorical values were filled using the most frequent category. Duplicate rows were removed to prevent repeated observations from biasing the results. The `last_updated` column was converted into datetime format to support time series analysis and date-based feature engineering.

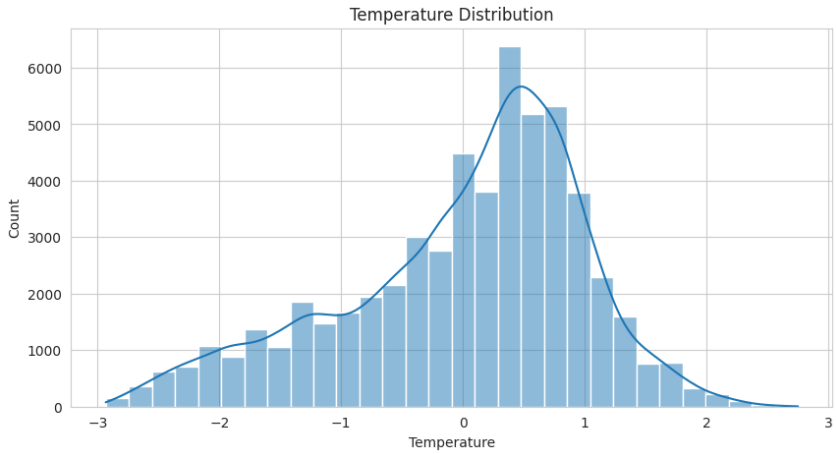
Outliers in numerical variables were treated using the Interquartile Range (IQR) method, which helps reduce the impact of extreme values. Finally, numerical features were standardized using `StandardScaler` to ensure consistent feature scales for machine learning models and improve model performance.

3.2 Exploratory Data Analysis (EDA)

A total of six exploratory visualizations were created to analyze distributions, temporal trends, correlations and geographical weather patterns. To keep the report concise and focused, the three most informative visualizations are discussed in detail below.

1. Temperature Distribution

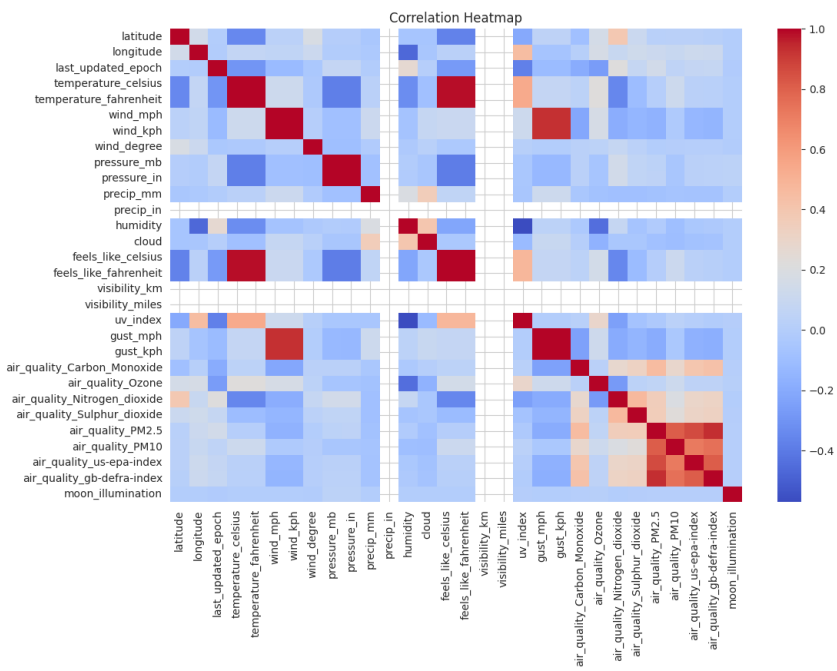
Figure 1: Temperature Distribution



The temperature distribution histogram shows that most observations are concentrated around moderate temperature values, with fewer occurrences at extreme cold and hot ranges. The slightly skewed shape reflects natural climatic variation across different countries and seasons.

2. Correlation Heatmap

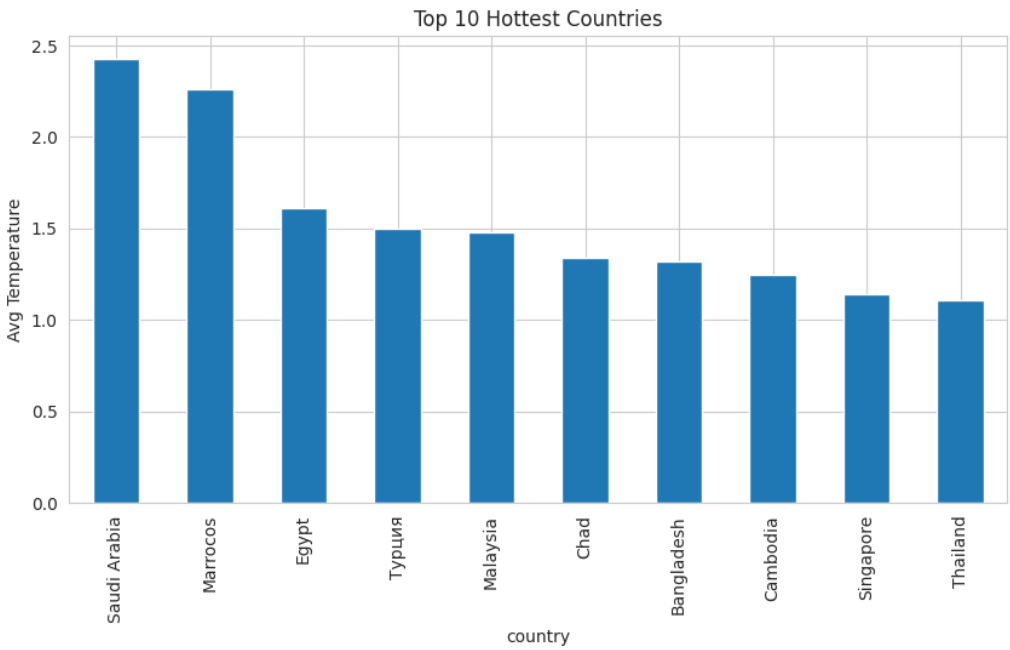
Figure 2: Correlation Heatmap



The correlation heatmap was used to study relationships among numerical weather variables. Strong positive relationships were observed between temperature and feels-like temperature, while several air quality indicators were also highly correlated. These findings helped identify useful predictors for forecasting models.

3. Top 10 Hottest Countries

Figure 3: Top 10 Hottest Countries



Country-level climate comparison showed that Saudi Arabia, Morocco and Egypt had the highest average temperatures in the dataset. This result aligns with expected climatic conditions and demonstrates the strong geographical diversity present in the data.

Additional visualizations such as precipitation distribution, rainfall trends over time and average temperature trends over time were also generated to support the broader exploratory analysis.

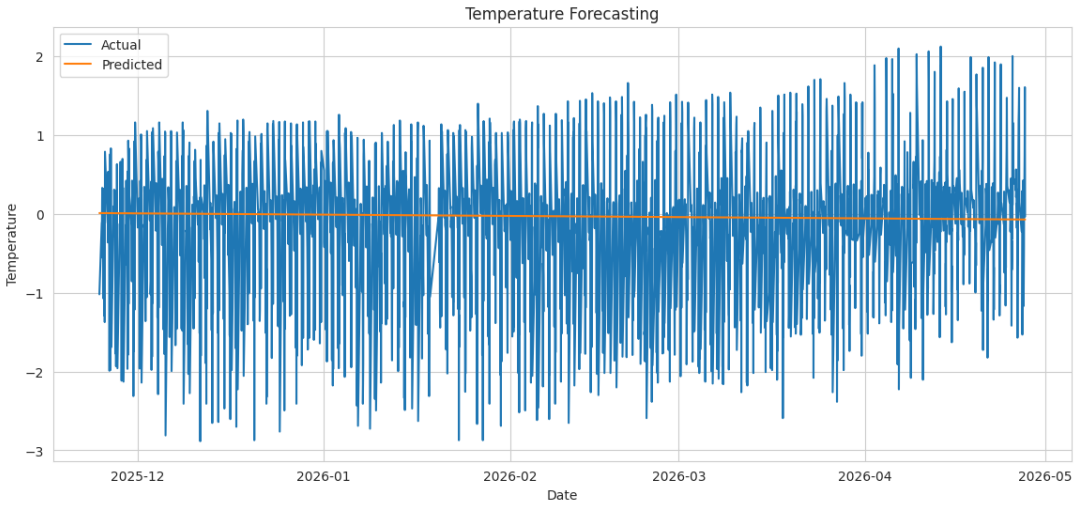
3.3 Forecasting Model Building

To establish an initial forecasting benchmark, a baseline Linear Regression model was developed using the `last_updated` timestamp feature. Daily average temperature values were first aggregated by date and the datetime variable was converted into a numerical day index to support regression-based time series prediction.

The dataset was divided chronologically into training and testing sets, where 80% of the observations were used for training and the remaining 20% were reserved for testing. This approach preserved the natural temporal order of the data and avoided future information leakage.

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). Although the Linear Regression model provided a simple baseline, its predictive performance was limited due to the nonlinear and highly dynamic nature of global weather patterns.

Figure 4: Baseline Temperature Forecasting using Linear Regression



The visualization above compares actual temperature values with predicted values on the test set. The results indicated that more advanced nonlinear models would likely provide stronger forecasting performance, motivating the next stage of model comparison and ensemble learning.

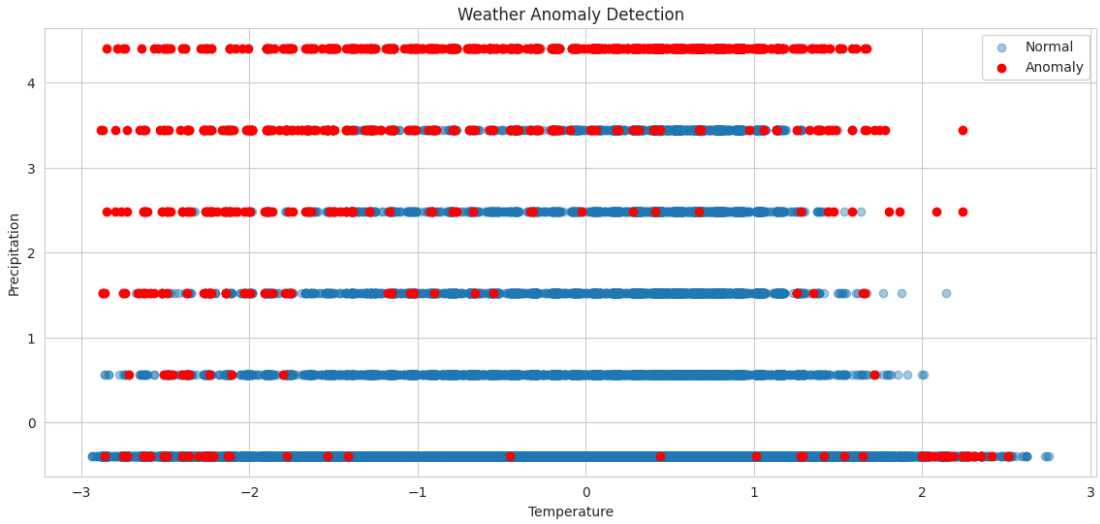
4 Advanced Assessment

4.1 Advanced EDA: Anomaly Detection

To identify unusual weather observations, an anomaly detection analysis was performed using the Isolation Forest algorithm. This unsupervised machine learning technique is effective for detecting rare patterns in large datasets without requiring labeled examples.

Five key weather variables were used for anomaly detection: temperature, humidity, wind speed, atmospheric pressure and precipitation. A contamination rate of 2% was selected, meaning the model classified a small portion of observations as potential anomalies.

Figure 5: Weather Anomaly Detection



The visualization above compares normal observations with anomalous records using temperature and precipitation values. Red points represent weather conditions that significantly differ from the majority of observations. These anomalies may indicate rare climate events, abrupt atmospheric changes or extreme environmental conditions.

This analysis provides additional value beyond forecasting by helping detect irregular weather behavior and improving understanding of unusual global climate patterns.

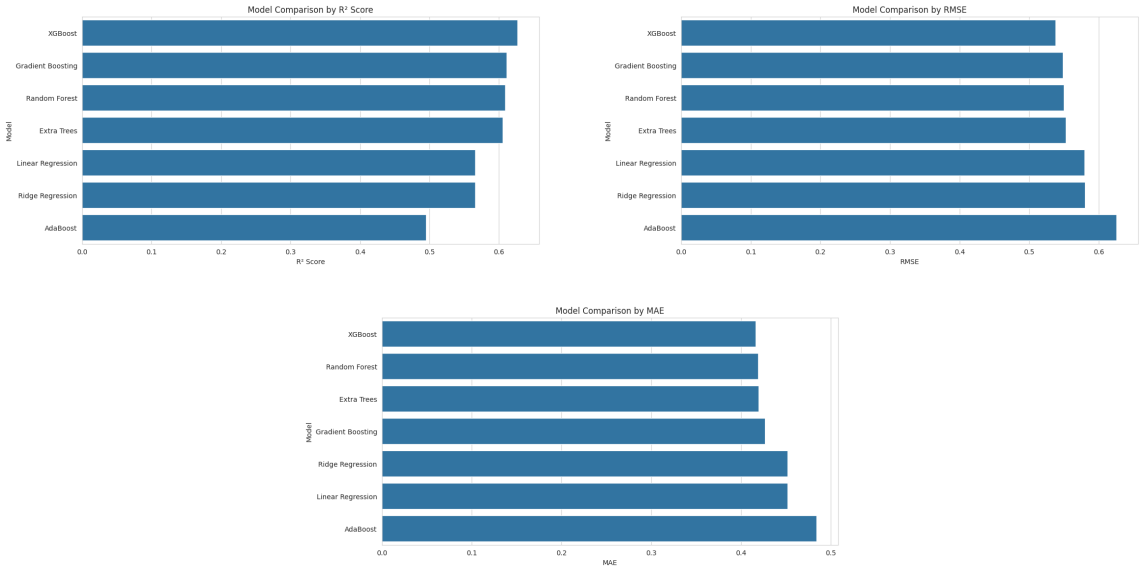
4.2 Forecasting with Multiple Models

To improve forecasting performance beyond the baseline model, several machine learning algorithms were trained and evaluated on the same training and testing dataset. The models compared in this study were Linear Regression, Ridge Regression, Random Forest, Gradient Boosting, Extra Trees, AdaBoost and XGBoost. Each model was trained using the prepared time-based and engineered weather features.

Model performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). Lower MAE and RMSE values indicate stronger predictive accuracy, while higher R^2 values indicate better overall model fit.

Among all standalone models, **XGBoost** achieved the best results with an MAE of **0.4163**, RMSE of **0.5377**, and R^2 score of **0.6264**. Gradient Boosting and Random Forest also performed strongly, with R^2 scores above 0.60. Linear models produced lower performance, while AdaBoost recorded the weakest results among the tested models.

Figure 6: Multiple Model Performance Comparison



Based on the model comparison results, the three strongest models were selected for ensemble learning: XGBoost, Gradient Boosting and Random Forest. A weighted ensemble model was created to combine the strengths of these algorithms and improve final predictive performance.

The final **Weighted Ensemble Model** achieved the best overall results with an MAE of **0.4139**, RMSE of **0.5344**, and R^2 score of **0.6310**. This outperformed every standalone model and demonstrated that combining high-performing models can increase forecasting accuracy and robustness.

The top three final performers were:

Table 1: Top 3 Performing Models

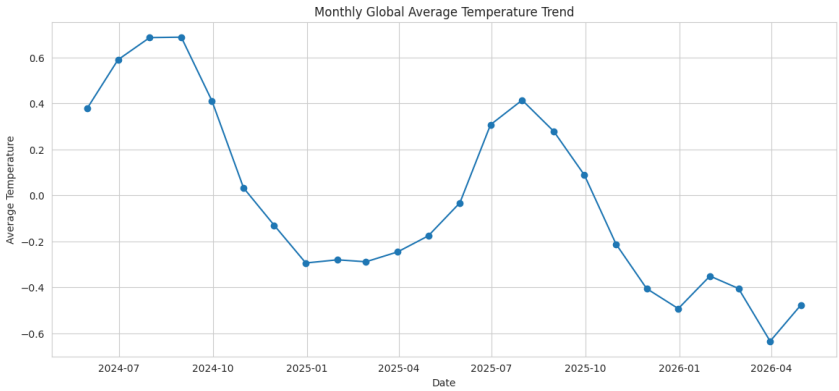
Model	R^2 Score
Weighted Ensemble	0.6310
XGBoost	0.6264
Gradient Boosting	0.6114

5 Unique Analyses

5.1 Climate Analysis

To explore long-term climate behavior, monthly global average temperature values were calculated and analyzed over time. This aggregation helps smooth daily fluctuations and reveals broader seasonal and temporal climate patterns across the dataset.

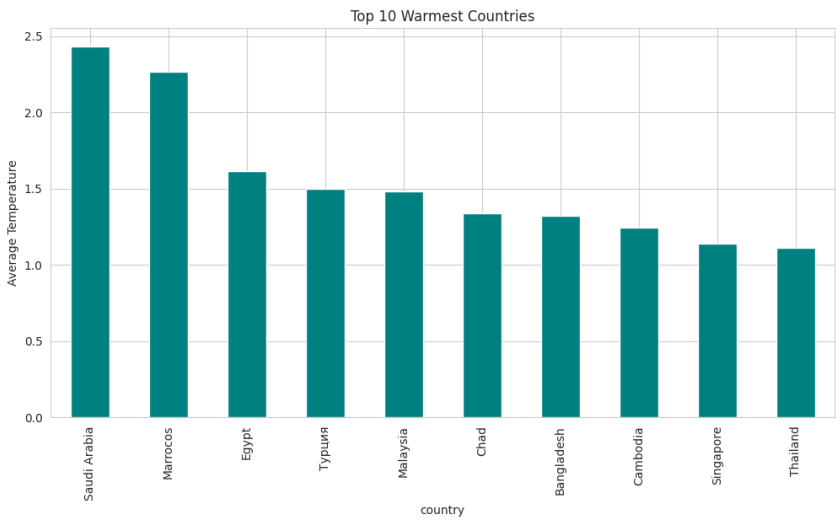
Figure 9: Monthly Global Average Temperature Trend



The monthly trend shows cyclical temperature movement, reflecting seasonal changes across different global regions. Warmer periods were followed by cooler intervals, indicating expected climate seasonality. Such trends confirm that time-based forecasting is appropriate for weather prediction tasks.

In addition to temporal analysis, country-level climate comparisons were performed to identify the warmest regions in the dataset.

Figure 10: Top 10 Warmest Countries



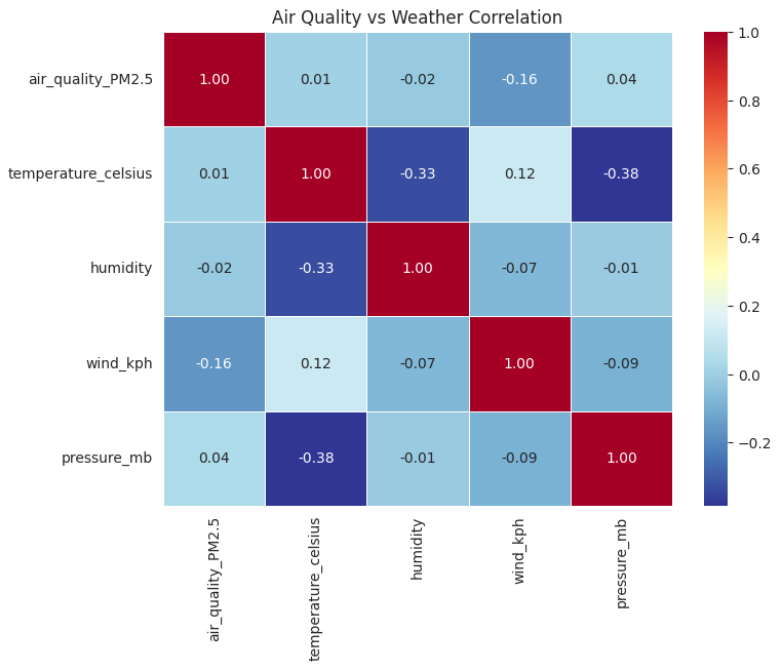
The results showed that Saudi Arabia, Morocco and Egypt recorded the highest average temperatures, followed by several tropical and arid countries. These findings align with known geographical climate conditions and demonstrate the global diversity captured within the dataset.

Overall, the climate analysis section highlights both temporal weather cycles and regional long-term temperature differences.

5.2 Environmental Impact Analysis

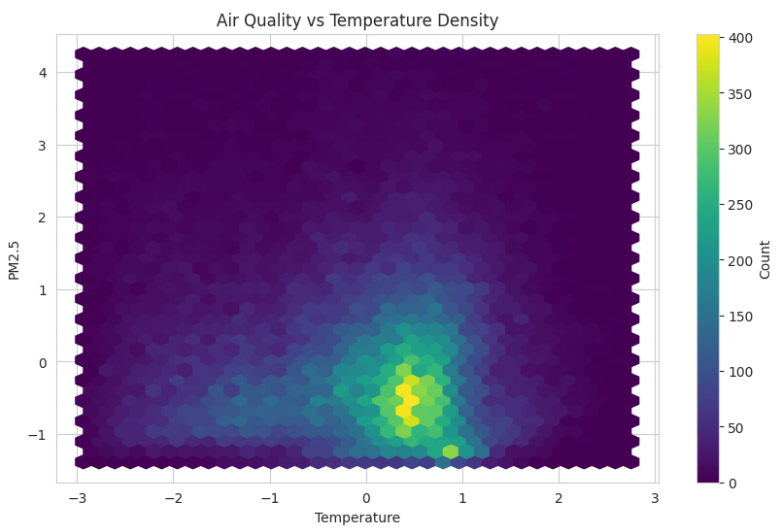
To study the relationship between environmental pollution and weather conditions, air quality indicators were analyzed alongside major atmospheric variables such as temperature, humidity, wind speed and pressure. PM2.5 concentration was used as the primary air quality metric due to its importance in public health and environmental monitoring.

Figure 11: Air Quality vs Weather Correlation



The correlation heatmap indicates that PM2.5 had weak positive relationships with temperature and pressure, while a slight negative relationship was observed with wind speed. This suggests that stronger winds may help disperse airborne particulate matter, reducing pollution concentration in some regions.

Figure 12: Air Quality vs Temperature Density



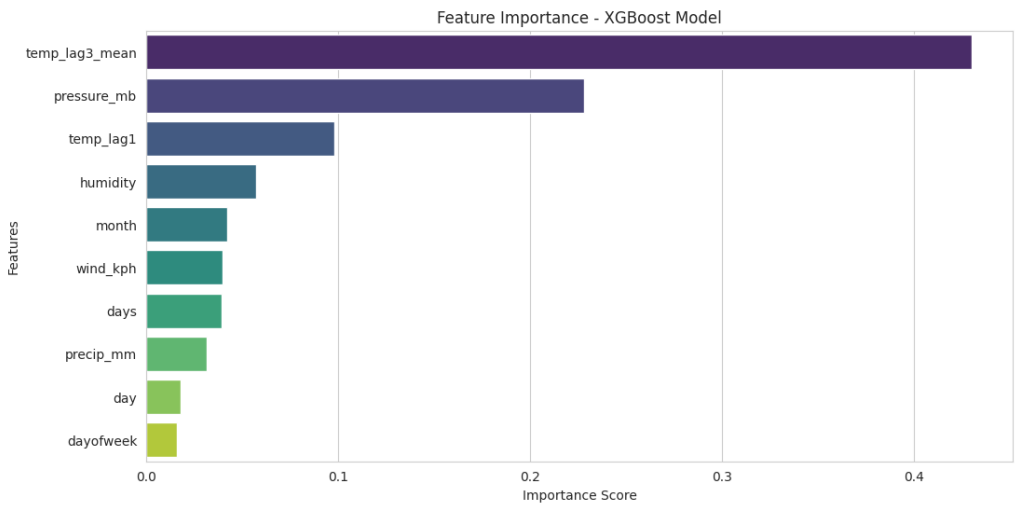
The density visualization further highlights where the largest concentration of observations occurred. Most data points were clustered around moderate temperature levels with relatively lower PM2.5 values, while fewer observations appeared in higher pollution ranges.

Overall, this analysis demonstrates that weather conditions can influence air quality behavior, although the relationships are moderate and may vary across locations.

5.3 Feature Importance Analysis

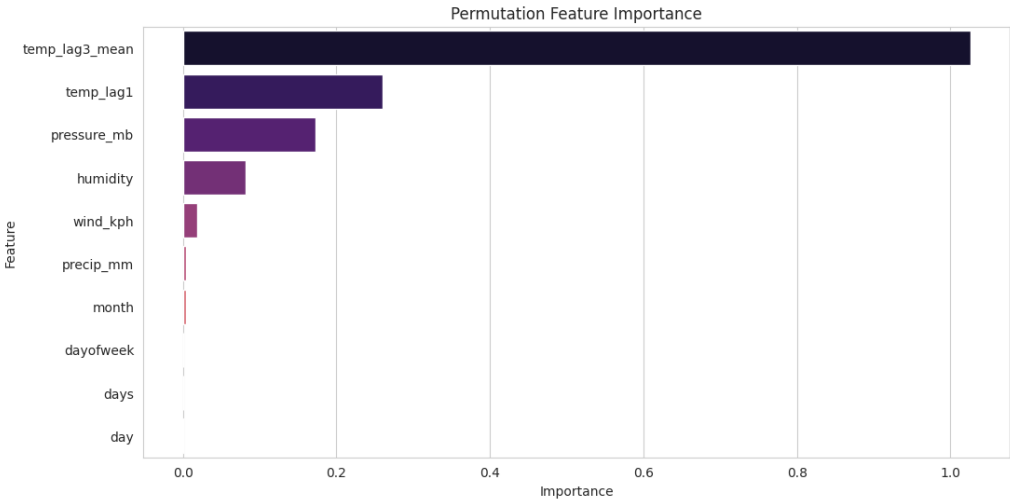
To better understand the drivers behind forecasting performance, feature importance analysis was conducted using the best-performing standalone model, XGBoost. Two complementary methods were used: built-in XGBoost importance scores and permutation importance.

Figure 13: XGBoost Feature Importance



The XGBoost importance plot shows that `temp_lag3_mean` was the most influential predictor, followed by atmospheric pressure and `temp_lag1`. This indicates that recent temperature history was the strongest contributor to future temperature forecasting.

Figure 14: Permutation Feature Importance



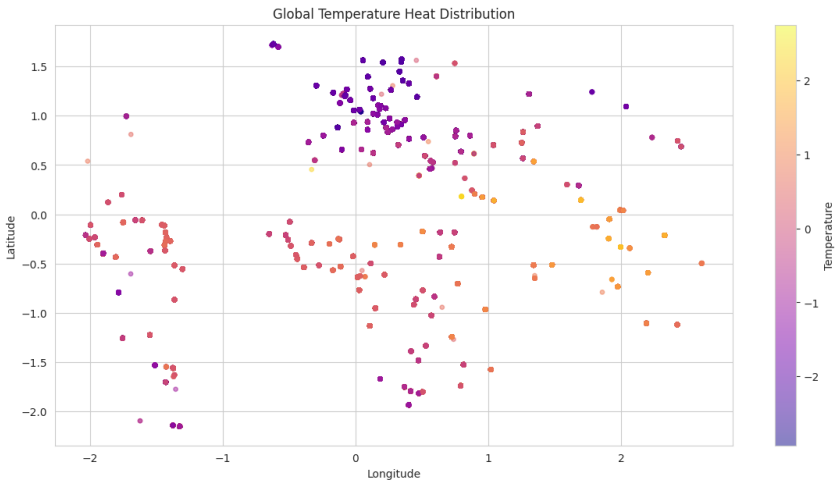
Permutation importance produced similar findings, again highlighting `temp_lag3_mean`, `temp_lag1`, and pressure as the most valuable features. The consistency between both methods increases confidence in the reliability of the model interpretation.

Overall, this analysis demonstrates that short-term historical temperature trends and atmospheric pressure were the most important variables driving model predictions.

5.4 Spatial Analysis

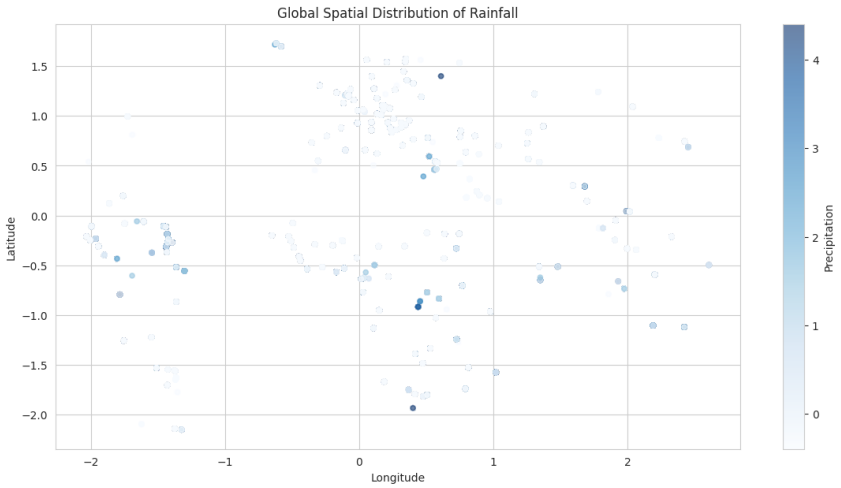
Geographical coordinates were used to visualize how weather conditions vary across different global regions. Spatial analysis helps reveal regional climate behavior and demonstrates the worldwide coverage of the dataset.

Figure 15: Global Temperature Heat Distribution



The temperature map shows clear geographical variation, with warmer observations concentrated in tropical and desert regions, while cooler temperatures were more common in higher-latitude areas. This aligns with expected global climate zones.

Figure 16: Global Spatial Distribution of Rainfall



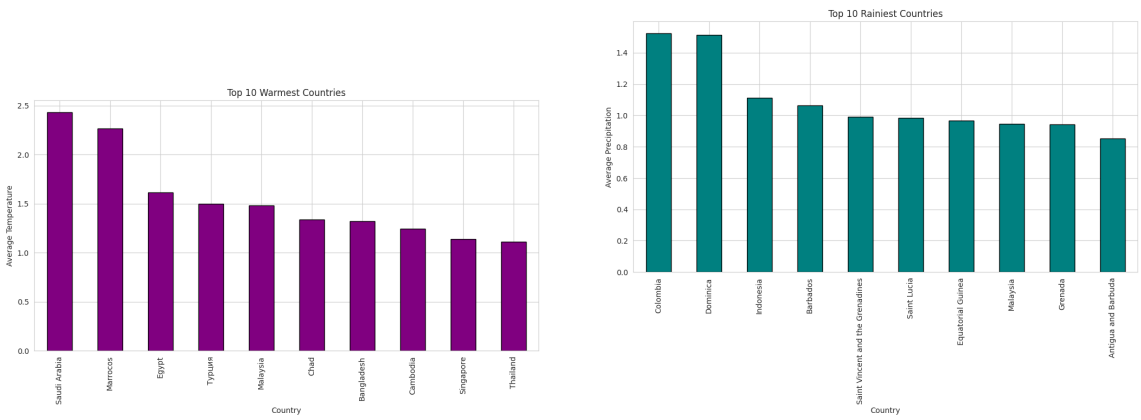
The rainfall distribution map indicates that precipitation levels were unevenly spread across locations. Certain tropical and coastal regions showed higher rainfall intensity, while arid regions experienced lower precipitation levels.

Overall, the spatial analysis confirms that both temperature and rainfall are strongly influenced by geography, making location an important factor in weather forecasting models.

5.5 Geographical Patterns

To further understand regional climate differences, country-level comparisons were performed using average temperature and precipitation values. This helps identify nations with extreme weather characteristics and highlights how climate varies across continents.

Figure 17: Regional Climate Comparison



The temperature comparison shows that Saudi Arabia, Morocco, and Egypt were among the warmest countries in the dataset, reflecting desert and arid climate conditions. In contrast, the rainfall comparison identified Colombia, Dominica, and Indonesia as some of the rainiest countries, consistent with tropical and coastal environments.

These findings demonstrate that geographical location strongly influences long-term weather behavior. Regions near deserts tend to record higher temperatures, while tropical and island nations generally experience greater rainfall levels.

6 Conclusion

This project successfully completed an end-to-end weather analytics and forecasting pipeline using a large global weather dataset. The workflow included data cleaning, preprocessing, exploratory data analysis, baseline forecasting, advanced machine learning model comparison, ensemble learning and multiple domain-specific analytical studies.

The forecasting results showed that tree-based ensemble models significantly outperformed traditional linear approaches. Among standalone models, XGBoost achieved the best performance, while the final weighted ensemble model delivered the highest overall accuracy with an R^2 score of 0.6310.

Additional analyses provided valuable insights into anomaly detection, air quality relationships, feature importance, climate trends and geographical weather variation. These findings demonstrate that weather data can support both predictive modeling and strategic environmental decision-making.

Overall, this assessment highlights strong practical skills in data science, machine learning, feature engineering, model evaluation and business-oriented analytical thinking. Future improvements may include deep learning time-series models, live API integration and deployment as an interactive forecasting dashboard.